Reg. No. | | | | | | | | | | | [12CE81]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

PAVEMENT ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.
3. Use of IRC code books permitted.

1. a. Compare the flexible and rigid pavements. Indicate why rigid pavements stand better. **07**

b. Explain the behavior of a flexible pavement and indicate why the number of layers are more in such pavements. **07**

2. a. What are the desirable properties of subgrade soil and indicate why clayey soil is not suitable as subgrade material. **07**

b. Plate load test was conducted on a soaked subgrade during rainy season on 30 cm diameter plate. And the following data was obtained. Find the modulus of subgrade reaction. **07**

Mean settlement, mm	0	0.24	0.52	0.76	1.02	1.23	1.53	1.76
Load values kg	0	460	900	1180	1360	1480	1590	1680

3. a. List and explain the various factors affecting flexible pavements. **07**

b. Explain the pavement design by IRC58:2011 method. **07**

4. a. With sketches explain the various joints in cement concrete pavements. **07**

b. Explain the effect of temperature on Rigid pavements and indicate how Bradbury calculated the temperature stresses using charts. **07**

5. a. Explain the different types of damages normally caused to flexible pavements and how they are rectified. **07**

b. Explain with a sketch the principle of BBD method and FWD method of pavement evaluation. **07**

6. a. Explain the overlay design procedure for flexible pavements based on IRC guidelines. **07**

b. Explain the principle of working of vibratory rollers, pneumatic tyred rollers and sheep foot rollers. **07**

7. a. Explain the principle of public private partnership in Transportation projects and what are the advantages of such projects. Quote an example for this concept. **08**

b. Write a note on benefit cost ratio of economic analysis. **06**

8. a. On an road accident happens due to a vehicle of 2t weight skidding to a distance of 40m and hits a parked vehicle of weight 1.0t and both the vehicles move together a distance of 12m before stopping. Assuming coefficient of friction of 0.5, find the initial speed of moving vehicle. **08**

b. Write the types of traffic signs usually used for roads and write one example for each with a neat sketch. **06**